

NATALIA KONG YI

Design Researcher · Service & Experience Designer · Systems Thinker

Translating research into strategy, and complexity into meaningful experiences

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The projects in this portfolio span healthcare, creative technology, and speculative design. Across these domains, they explore a common question: how can design help people understand, navigate, and shape complex systems?

My practice combines design research, systems thinking, and experience design to investigate how technology, services, and physical environments influence human behavior and collective outcomes.

Rather than treating products as isolated interfaces, I design interventions that strengthen the relationships between people, organizations, and living systems.

PRACTICE



Beyond the Clinic. *Conceptual framework used to understand the relationships between patients, knowledge, community, and care services before defining individual interventions.*

DIABETES INTEGRAL HEALTHCARE CENTER

Designing connected care experiences across fragmented healthcare services

Role: Impact Designer · Service Design Team

Year: 2022

Type: Service Design · Healthcare Innovation · Participatory Design

Team: Viernes Peru · Multidisciplinary design team

CHALLENGE

Diabetes management extends beyond clinical care, reshaping everyday routines, relationships, and long-term lifestyle change. Yet most healthcare services remain focused on medical treatment rather than the broader support systems that sustain it.

How can healthcare services support the everyday relationships and practices that influence long-term treatment engagement?

Through participatory research, the team explored how care could extend beyond clinical interactions by designing services, tools, and experiences that foster empathy, dialogue, and shared responsibility among patients and their support networks.



Ease
Burden



Share
Experiences



Encourage
Reflection



Actionable
Knowledge



Spark
Curiosity

Learning principles. *Design principles used to guide every learning artifact and experience across the project.*

OUTCOMES

Developed a service ecosystem integrating research insights, interaction tools, and strategic frameworks to support long-term diabetes care.

Care Ecosystem

Reframed diabetes management as an ecosystem shaped by relationships, routines, and everyday contexts rather than clinical interactions alone.

Shared Practices

Designed and prototyped interaction tools—including conversation cards and an interactive installation—to encourage empathy and collaborative care. Experience prototype evaluated with 10 participants.

Service Strategy

Created a design playbook translating research insights into reusable principles for future healthcare services.

Learning Artifact Design Canvas

Subject  1 What topic are we exploring? What core idea do we want to develop? How does this idea contribute to the learning and community ecosystem?	Objectives:  2 How does this topic support the strategic goals of the healthcare center? What outcomes are we aiming to achieve? Who is this experience designed for? What insights or outputs do we hope to generate?	Design & Fabrication:  7 How will this artifact be designed? What are its key components? What format will it take? Does it require any supporting elements for implementation (furniture, lighting or acoustic systems, audiovisual equipment, or other resources)?
Relational system:  4 How does this artifact connect with the broader learning ecosystem? What other learning artifacts or community activities does it relate to? What upcoming community events should it support or build upon?	Approach:  3 How should this topic be approached? In what context will this artifact be experienced (public, semi-private, or private)? What level of participation or visibility should it encourage?	
Principles:  5 How should the learning principles be reflected in this artifact? Rate the relevance of each principle and indicate its importance using the corresponding scale.	Users:  6 Who is this artifact designed for? What experience, behavior, or outcome do we want to encourage for each audience?	

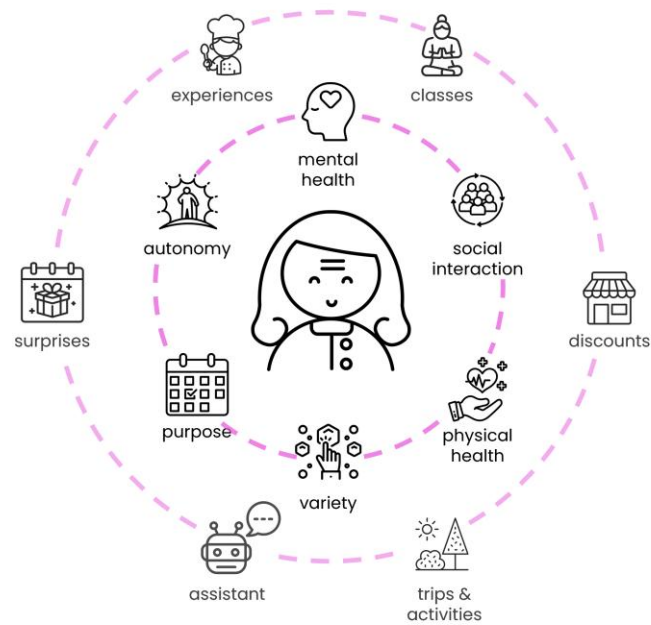
Learning Artifact Design Canvas. *Framework for translating strategic objectives into cohesive learning experiences.*

TAKEAWAYS

- Care extends beyond the clinic. Health outcomes are shaped by relationships, routines, and everyday environments as much as by clinical interventions.
- Relationships are part of the care system. Designing opportunities for dialogue and shared reflection can strengthen long-term treatment engagement.
- Design frameworks enable continuity beyond individual interventions. Translating research into reusable principles enables teams to design, evaluate, and evolve services beyond a single intervention.

Research & Design: 6 stakeholder interviews · Service ecosystem mapping · Co-design workshops · MVP validated with 10 patients

Outputs: Ecosystem map · Learning Playbook · Family Playbook · Design canvases



Seniors Club Ecosystem. Mapping multi-stakeholder participation between seniors, families, and specialized service providers.

SENIORS CLUB

Designing service ecosystems that strengthen autonomy, belonging, and community in later life

Role: Co-founder · Strategy & Service Designer

Type: Service Strategy & Concept

Year: 2025

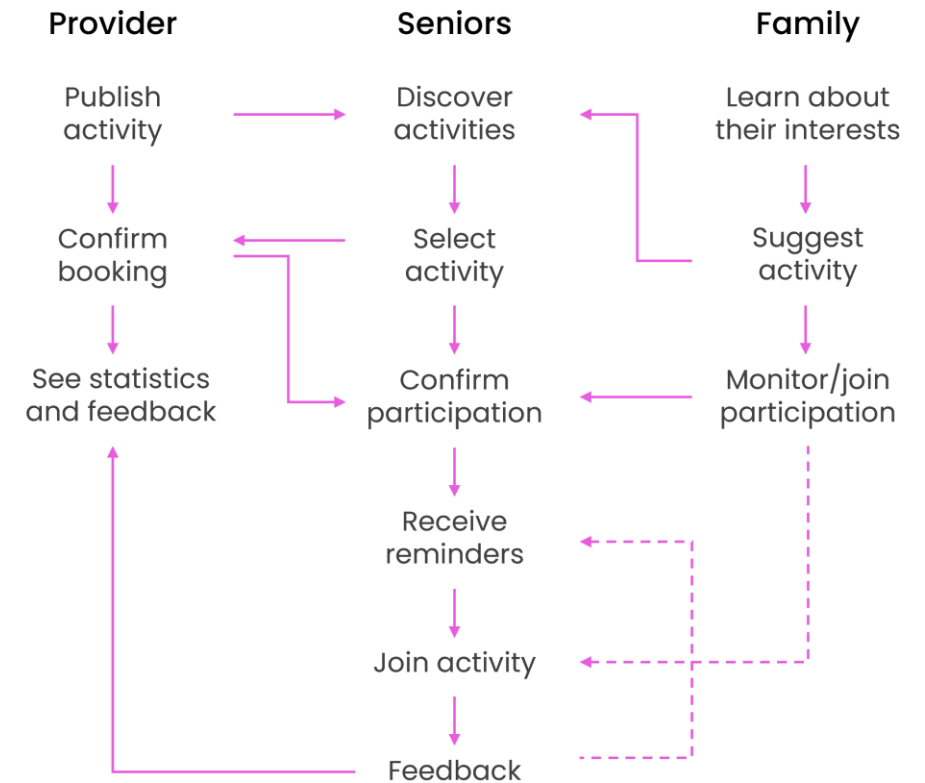
Team: Co-founded initiative

CHALLENGE

As populations age, maintaining independence depends not only on healthcare, but also on opportunities for participation, purpose, and social connection. Yet many services focus on assistance rather than enabling active aging.

How might a service ecosystem help older adults remain connected, independent, and engaged while supporting families without reducing autonomy?

We explored this question through a strategic service concept combining research, ecosystem thinking, and experience design. The proposal examined how partnerships, digital coordination, and physical touchpoints could foster long-term belonging rather than isolated service interactions.



Platform Flow. Role-based interactions defining the coordination and feedback loops between seniors, families, and providers.

OUTCOMES

Explored how ecosystem design, experience architecture, and partnership models could support long-term participation and independent living.

Service Ecosystem

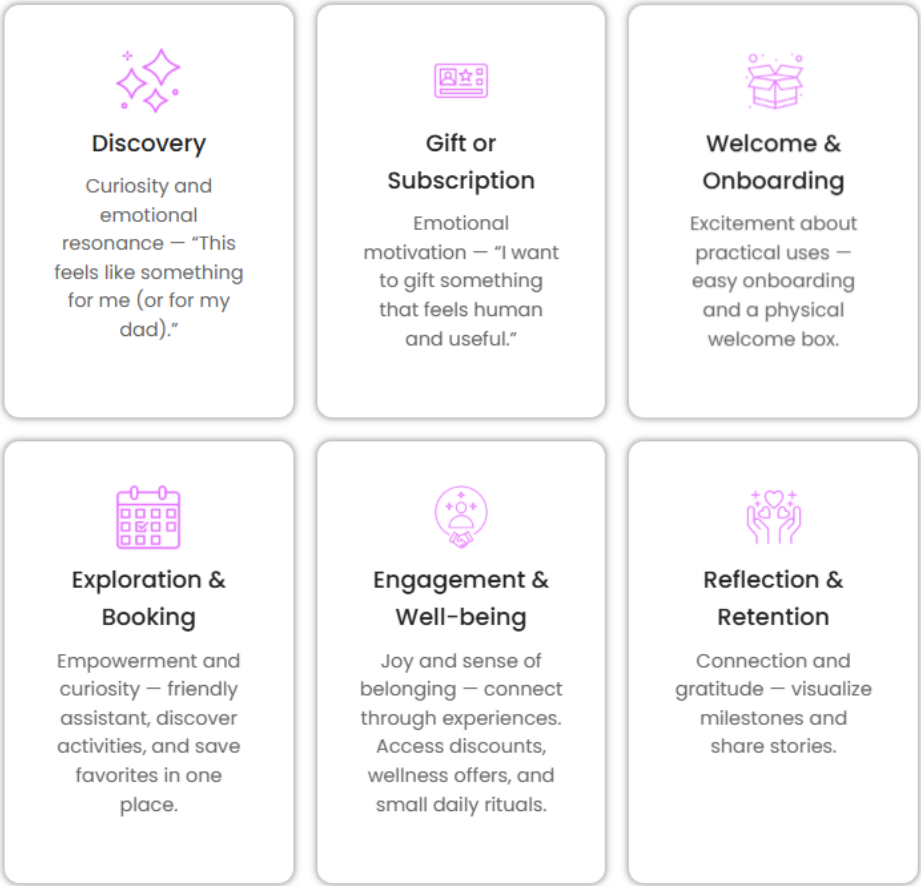
Defined an end-to-end service model connecting older adults, families, and partner organizations through coordinated experiences.

Experience Framework

Developed the service journey, interaction principles, and touchpoints supporting participation from onboarding to long-term engagement.

Strategic Concept

Proposed the value proposition, partnership model, and service architecture as a foundation for future prototyping and validation.



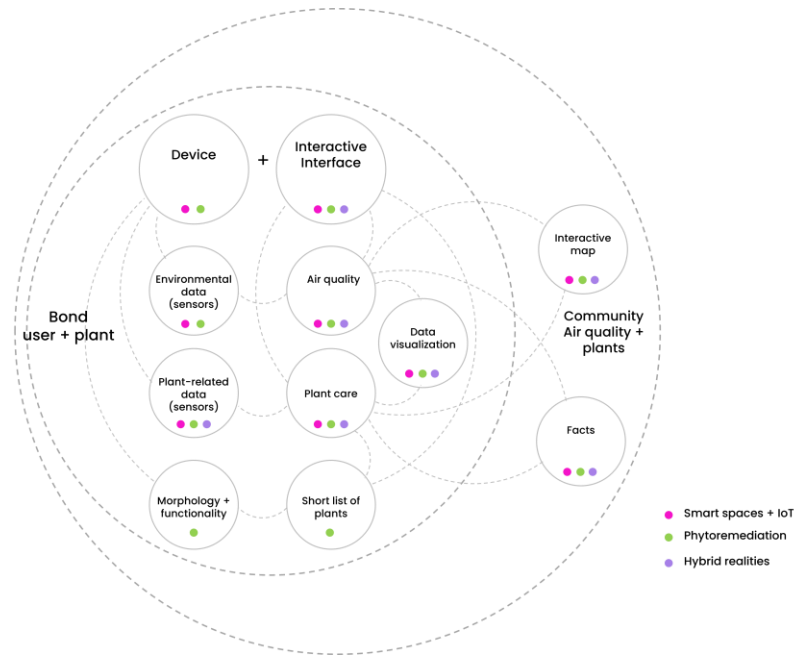
Experience Lifecycle. *Defining the emotional and functional touchpoints of the user journey.*

TAKEAWAYS

- Care extends beyond healthcare. Well-being depends on relationships, participation, and everyday experiences.
- Agency encourages participation. Services should reinforce autonomy rather than dependence.
- Communities become care infrastructures. Social networks and local organizations strengthen long-term well-being.
- Continuity builds belonging. Services should evolve alongside people's changing needs.
- Strategy precedes implementation. Clarifying relationships and value exchanges creates stronger foundations for future services.

Methods: Service blueprinting · Journey mapping · Ecosystem mapping · Value proposition design

More: [Full case study](#) →



Conceptual system map. *Relationships between environmental sensing, digital representation, and user interaction, used to align the platform's architectural logic with the intended user experience.*

HYPERPLANT

Designing hybrid experiences that cultivate long-term relationships between people and living systems

Role: Interaction & Systems Designer · Creative Technologist

Type: Master's Thesis · Research through Design

Year: 2024

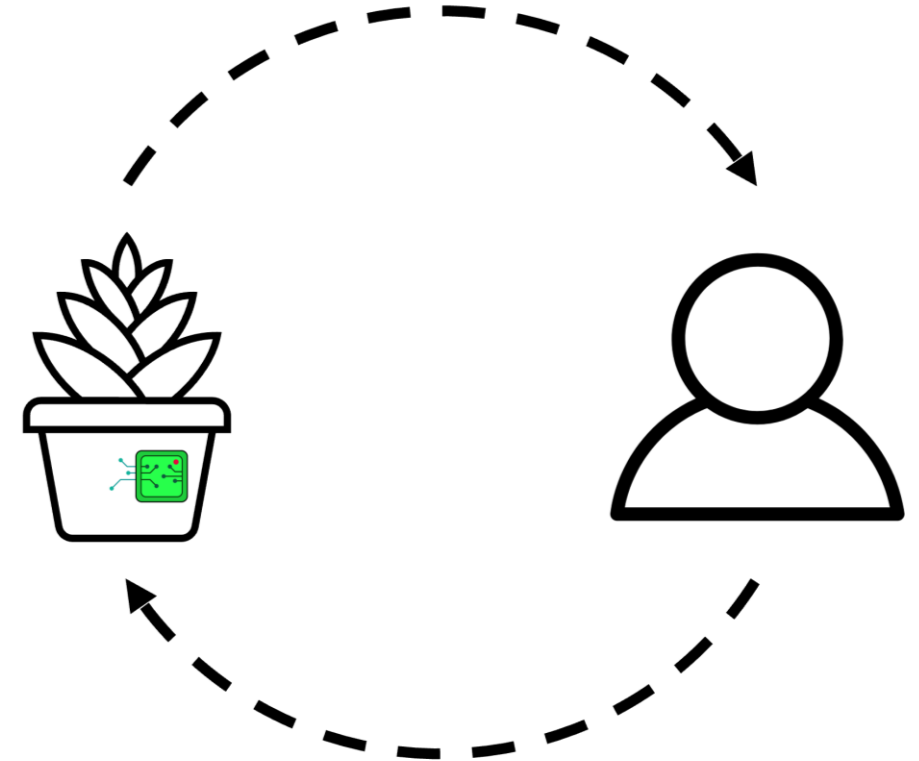
Team: Individual Project

CHALLENGE

Current models of production and consumption often separate people from the ecological systems that sustain them. Although environmental data can make these processes visible, it rarely fosters lasting awareness or behavioral change.

How might hybrid interactive systems cultivate care and responsibility toward living systems?

Research-through-design project that aims to reframe human-nature relationship through interaction and data. HyperPlant integrates living digital twins, environmental sensing, and interactive feedback into a hybrid ecosystem that extends human perception beyond observation toward ongoing engagement.



Conceptual Framework. Defining the platform architecture and the relationship between sensing and user interaction.

OUTCOMES

Developed and evaluated a hybrid system combining IoT, interaction, and behavioral design to explore long-term engagement with living systems.

Hybrid Interaction Ecosystem

Integrated environmental sensing, physical computing, and a web platform into a connected physical–digital experience.

Living Digital Twins

Translated environmental conditions into expressive digital representations that made ecological change perceptible through everyday interaction.

Smart Plant Wearable

Designed and prototyped a 3D-printed wearable that monitors environmental conditions while supporting air phytoremediation.

Community Platform

Proposed a digital platform for monitoring, storytelling, and future community participation.

Behavioral Evaluation

Conducted a one-week user evaluation using participant diaries and the Octalysis framework to examine engagement and motivational patterns.



System Components. The underlying components required to capture, process, and transmit environmental data.

TAKEAWAYS

- Invisible systems become meaningful through interaction. Design can translate imperceptible ecological processes into everyday experiences.
- Reciprocity strengthens care. Making mutual influence visible fosters more meaningful relationships with living systems.
- Hybrid experiences connect perception and action. Physical and digital interactions reinforce one another when grounded in the same reality.
- Narrative sustains engagement. Stories and feedback turn awareness into sustained action.
- Embodied technologies expand perception. Sensors, computation, and biology create new ways of perceiving and interacting with the living world and environment.

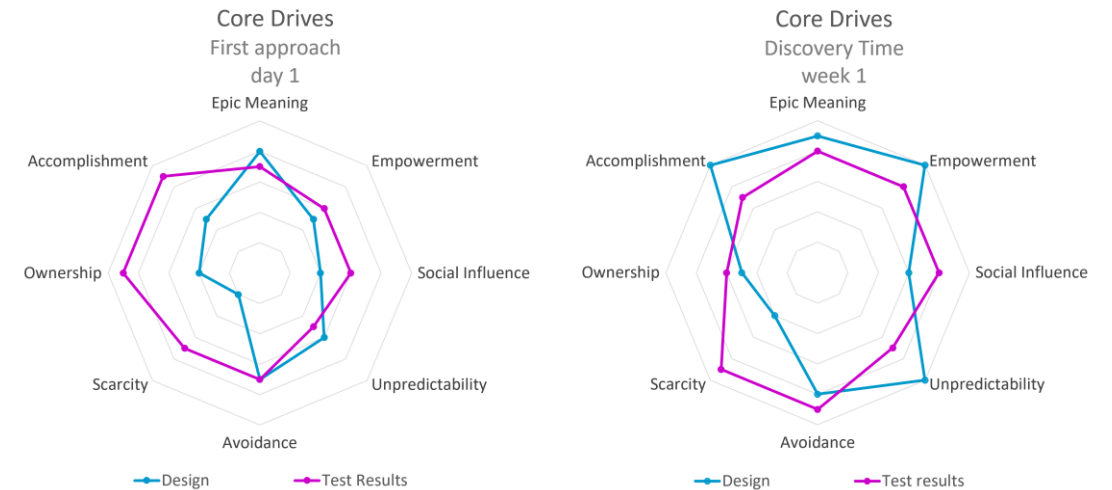
Methods: Research-through-design · Comparative research · Physical prototyping · AI-assisted coding · one-week user evaluation · Octalysis analysis

Tools: Arduino · Sensors · Physical computing · 3D printing · Interactive prototyping

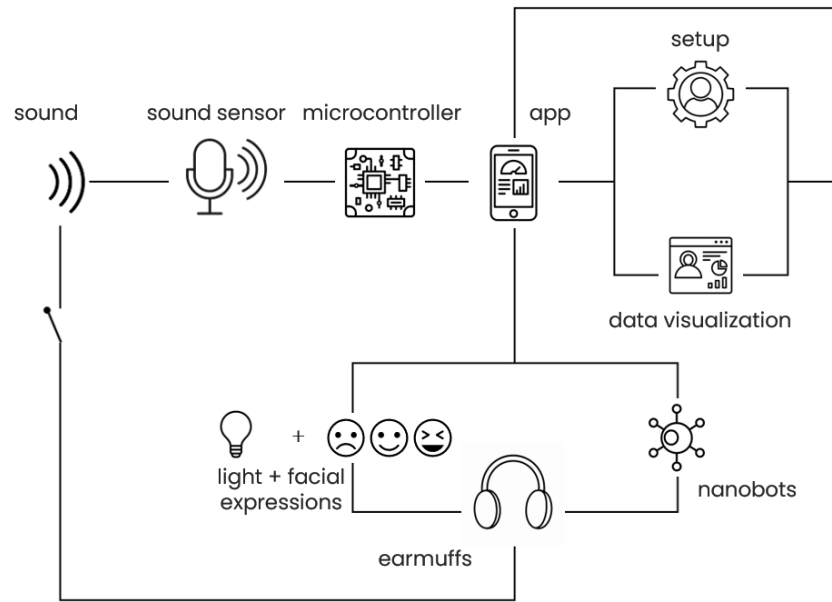
Presented at: Chinese Conference on Bio-Inspired Design and Technology (2020)

· International Image Festival (Colombia, 2021)

More: [Full case study](#) →



Behavioral Analysis. Using the Octalysis framework to measure the alignment between designed motivation models and real user engagement.



Nanomuffs system. *Environmental comfort layer connecting a wearable interface with data-driven interaction..*

NANOMUFFS

Imagining wearable technologies for empathy in a future shaped by altered sensory perception

Role: Speculative Designer · Creative Technologist

Type: Speculative Design · Design Fiction · Wearable Concept

Year: 2025

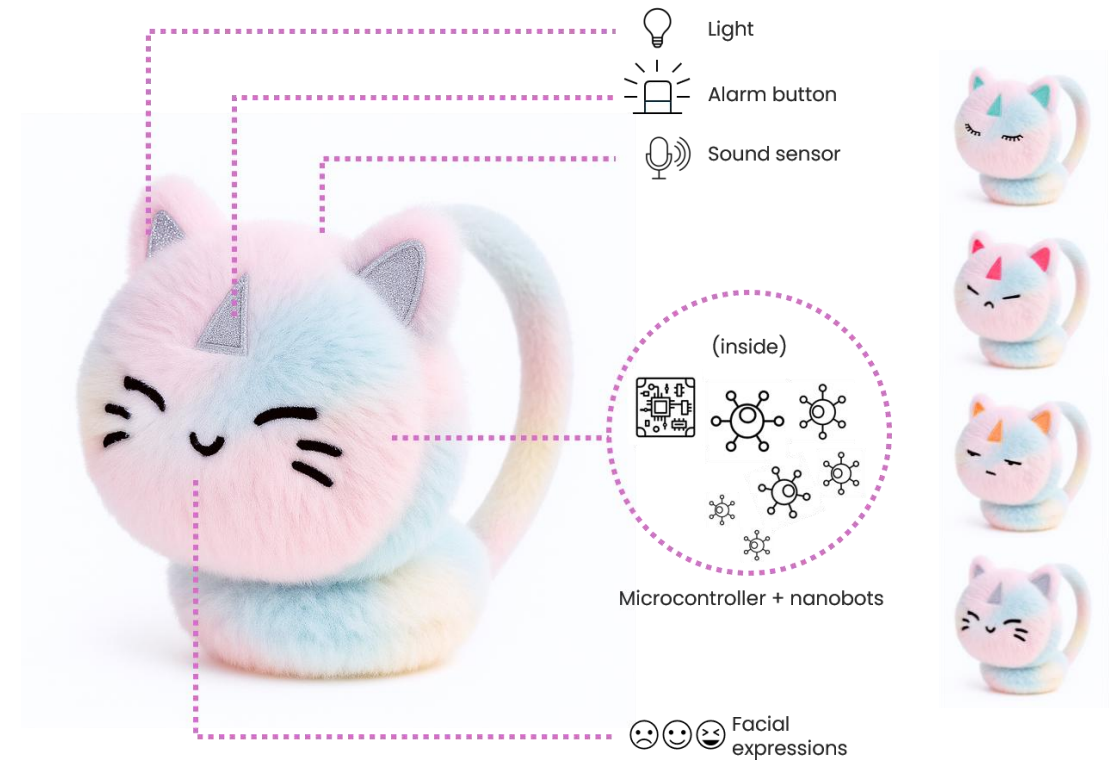
Team: Individual academic project

CHALLENGE

As biotechnology becomes increasingly capable of modifying human biology, future societies may need to navigate not only new abilities, but also new forms of sensory difference. Nanomuffs imagines a dystopian future in which widespread genetic interventions have resulted in heightened auditory sensitivity among younger generations, creating new challenges for communication and social interaction.

How might community-centered design foster empathy and facilitate interaction in a world shaped by amplified sensory perception?

The project explored this question through speculative prototyping, imagining a wearable device that makes auditory comfort visible to others. The concept connects biological, digital, and emotional dimensions of care, using expressive interfaces to communicate an otherwise invisible sensory state.



Nanomuffs components and interface. *Perceive the user's sound-comfort level through the Nanomuffs face. Magnetic ink defines the eyes, LED lights indicate auditory balance through the ears, and an emergency button on the forehead blocks all sound when activated.*

OUTCOMES

The project developed a speculative scenario and wearable concept to explore how future technologies might make sensory differences more visible and support more empathetic forms of interaction.

Speculative Wearable Concept

Developed Nanomuffs, an imagined wearable system that communicates the user's auditory comfort through expressive visual cues.

Sensory Interface

Explored a conceptual interface combining facial expressions, LED indicators, and an emergency sound-blocking function to make an otherwise invisible sensory state legible to others.

Alternative Future

Used design fiction to question technology narratives centered on optimization and imagine biotechnology as a potential mediator of empathy, inclusion, and social connection.

TAKEAWAYS

- Invisible experiences can become interaction layers. Making sensory states perceptible can create opportunities for greater mutual understanding.
- Speculation can challenge technology narratives. Imagining alternative futures helps question whether emerging technologies should optimize people or support different ways of being.
- Empathy can be designed into systems. The project explored technology not only as a functional tool, but as a mediator of relationships and social awareness.

Methods: Design fiction · Speculative design · Future Wheel · Scenario building

Scope: Speculative exploration in futures design developed as part of the Master in Creative Computing at ELISAVA / UVic-UCC.

More: [Full case study →](#)



Water Flow Series. *Exploring rivers as living data streams, where environmental conditions become visual narratives.*

HYBRID NATURES

Designing participatory infrastructures for engaging with living systems through hybrid experiences.

Role: Strategy & Systems Designer · Design Research · Creative Technologist

Year: 2025

Type: Master's Thesis (Creative Computing, ELISAVA / UVic-UCC)

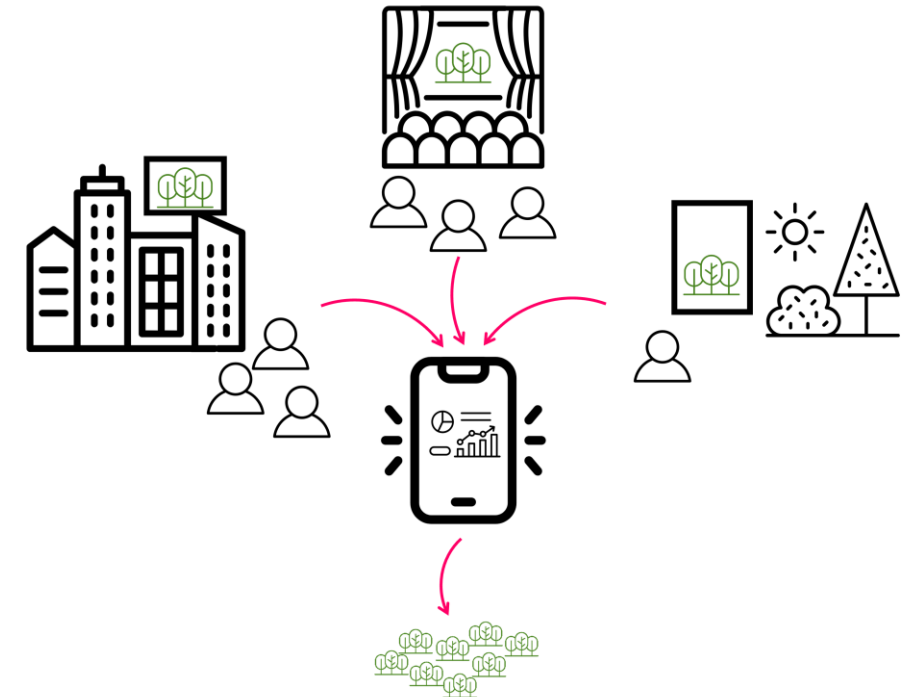
Team: Independent project

CHALLENGE

Hybrid Natures is a research-through-design project exploring how hybrid physical–digital systems can help people understand and engage with complex socio-environmental systems. It investigates how data, narrative, and interaction can make distant ecosystems perceptible through participatory experiences.

How can design and technology make distant human–nature relationships legible and actionable at a human scale?

Rather than designing a standalone product, I approached the challenge as a systems problem, integrating interaction, governance, and participation into a single experience.



Hybrid Natures Concept. Conceptual framework connecting environmental data, digital ownership, and collective stewardship through a decentralized platform.

OUTCOMES

Developed a proof of concept integrating interaction design, systems thinking, and technology, to explore new forms of collective environmental stewardship.

Narrative

Designed a data-driven experience that guides participation from awareness to contribution, explaining the problem, the impact and including a call to action.

Participation

Integrated governance, digital ownership, and collective decision-making into a participatory system concept.

Value Model

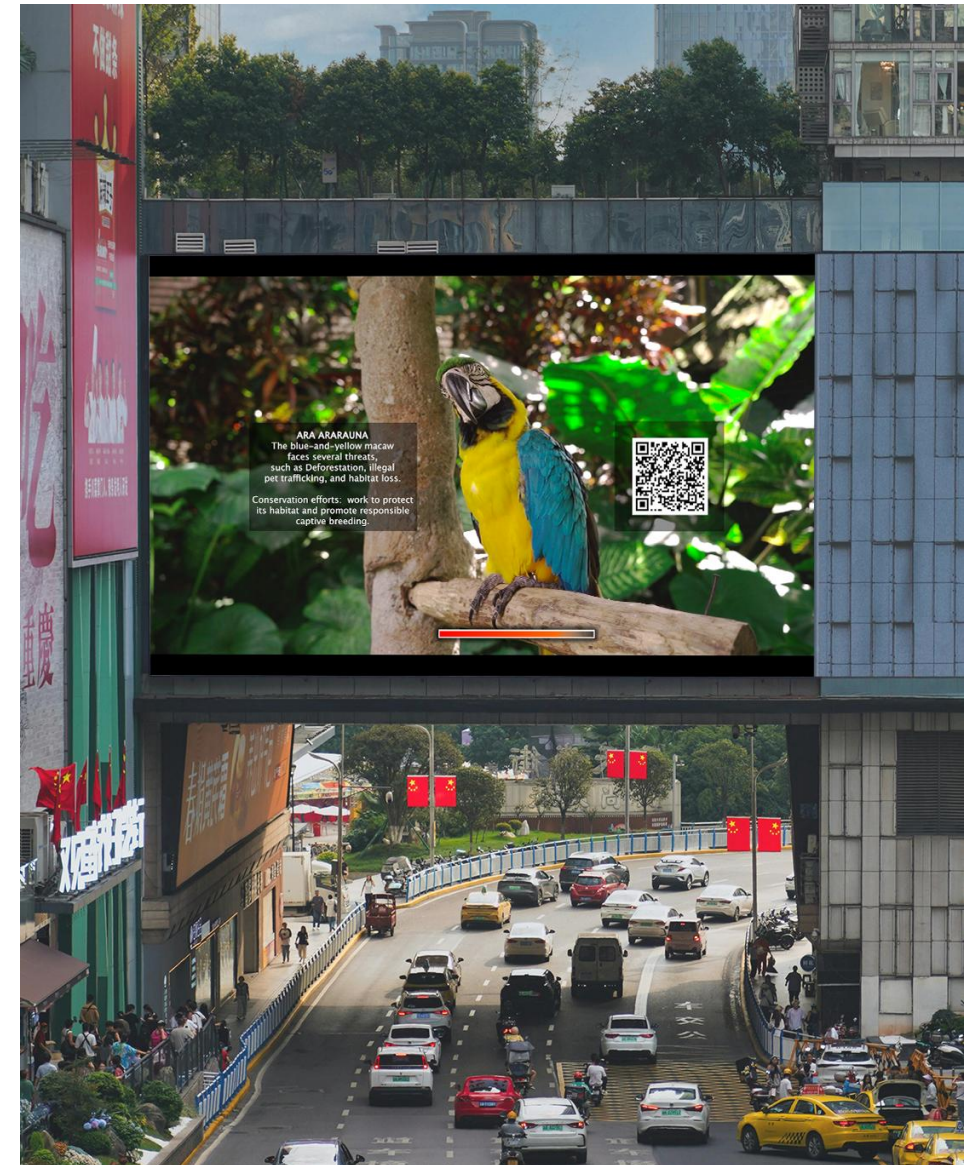
Proposed a shared-value model allocating variable revenue between artists and conservation NGOs, to align economic and environmental incentives.

Hybrid Infrastructure

Defined a platform architecture connecting digital participation with distributed decentralized physical touchpoints.

Creative Technology

Built a functional prototype using AI-assisted development, generative visualization, and web technologies.



Decentralized screens (speculative). *Interface concepts for discovering environmental narratives and supporting conservation through digital ownership.*

TAKEAWAYS

- Design participation intentionally. Long-term engagement emerges when ownership, contribution, and progression become part of the experience.
- Governance shapes the product experience. Decision-making and transparency influence how people engage—not only how systems operate.
- Narrative makes complexity navigable. Structuring information across multiple scales helps people build mental models of interconnected systems.
- Hybrid systems extend engagement. Physical and digital touchpoints reinforce one another, creating richer and more persistent participation.
- Speculation as a design method. Designing and prototyping future scenarios makes assumptions tangible before full implementation.

Tools: p5.js · HTML/CSS · Flourish · AI-assisted development

Presented at: International Colloquium Extractivismos (Argentina, 2025)

More: [Full case study](#) →

Education

2024-2025 | Master in Creative Computing. ELISAVA & University of Vic – Central University of Catalonia.

2018-2019 | M. Des in Interactive Design. University of Buenos Aires. (Thesis defended in 2024)

2010 | B. Arch. Pontifical Catholic University of Peru

Research + Exploration Areas

Hybrid Experiences · Systems · Creative Computing · Participatory Models · Generative Design

Publications, Conferences and Lectures

2025 | International Colloquium ExtrActivismos, National University of San Martin. Project presentation (Hybrid Natures).

2022 | Design, Art and Science chair, University of Bogota Jorge Tadeo Lozano. Guest lecture & project presentation (HyperPlant).

2021 | EXPERIENCE DESIGN Korea & Latin America Research Exchange. Human and Design Press. Seoul (KR). Paper co-author (published).

2021 | International Image Festival (CO). Project presentation (HyperPlant) 2020 | Chinese Conference on Bio-inspired Design and Technology (CN). Project presentation (HyperPlant)

Selected Projects

2026 | Kuya Silver Mining Facilities — Architectural Consultant

2026 | Smart Cargo — Strategy Designer

2025 | Seniors Club — Strategy Designer

2025 | Zapallal Mall — Architectural Consultant

2024 | Talara Mall — Architectural Consultant

2024 | HorrorLand — Experience Designer (Immersive Systems)

2023 | Circurela — Designer (Pop-up & Circular Systems)

2022 | Diabetes Integrated Healthcare Center — Experience Designer

2022 | La Molina Mall — Architectural Consultant

2021 | AskMe — Design Director

2019 | S'Dely Corporate Building & Showroom — Principal Architect

2018 | MegaPlaza Resettlement — Architectural Consultant

2017 | Osinergmin — Design Standards Handbooks (National Scale)

2016 | MegaPlaza — Design Standards Handbooks (National Scale)

2015 | VeggieLab — Principal Architect

2014 | Onduline — Marketing & Technical Department Manager

2010 | Participatory Urban Design Workshop — Urban Consultant

EXPERIENCE